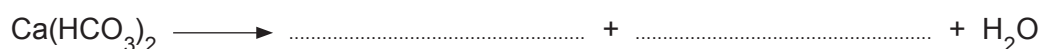


6. Water treatment is an important process in producing water for drinking. Temporary hard water contains calcium hydrogen carbonate $\text{Ca}(\text{HCO}_3)_2$ and magnesium hydrogen carbonate $\text{Mg}(\text{HCO}_3)_2$. Boiling breaks down the hydrogen carbonate to form insoluble carbonates. It is an inexpensive method used to soften temporary hard water.

- (a) (i) Complete the equation below to show this process. [2]



- (ii) Explain why permanent hard water cannot be softened by boiling. [2]

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- (b) The addition of sodium carbonate or passing water through an ion exchange column can be used to soften temporary and permanent hard water.

Explain how the following methods soften water.

- (i) Addition of sodium carbonate. [2]

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- (ii) Passing water through an ion exchange column. [2]

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- (c) State **one** disadvantage other than cost associated with an ion exchange column. [1]

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- (d) Describe the effects of hard water on the central heating system in a house. [2]

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